

Docker Commands & Flags – HandsOn

1. Docker Basics

Check Docker version

```
docker version
```

- Client + Server version
- Confirms Docker daemon connectivity

```
docker info
```

- System-wide info: storage driver, cgroups, images, containers

2. Image Management

List local images

```
docker images
```

Flags:

- `-a` → show all images (including intermediate layers)
- `-q` → only image IDs

Pull image from registry

```
docker pull ubuntu
```

```
docker pull ubuntu:22.04
```

- Downloads image from Docker Hub (default registry)
- `:tag` specifies version

Remove image

```
docker rmi ubuntu
```

Flags:

- `-f` → force removal
- `-a` → remove all unused images

3. Container Lifecycle

Run a container

```
docker run ubuntu
```

Common flags:

- `-it` → interactive + terminal
- `--name mycontainer` → custom container name
- `-d` → detached (background)
- `--rm` → auto-remove container after exit

Example:

```
docker run -it --name test ubuntu bash
```

List containers

```
docker ps
```

Flags:

- `-a` → all containers (stopped + running)
- `-q` → only container IDs

Start / Stop / Restart

```
docker start container_name  
docker stop container_name  
docker restart container_name
```

Remove container

```
docker rm container_name
```

Flags:

- `-f` → force stop and remove
- `-a` (with `prune`) → remove all stopped containers

4. Executing Commands Inside Containers

Attach terminal

```
docker attach container_name
```

Execute command in running container

```
docker exec -it container_name bash
```

Flags:

- `-i` → interactive
- `-t` → terminal

5. Networking & Ports

Port mapping

```
docker run -d -p 8080:80 nginx
```

Explanation:

- Host port
- Container port

Flags:

-
- → random host ports

List networks

```
docker network ls
```

Create network

```
docker network create mynet
```

Attach container:

```
docker run -d --network=mynet nginx
```

6. Volumes & Data Persistence

Create volume

```
docker volume create mydata
```

Mount volume

```
docker run -v mydata:/data ubuntu
```

Bind mount:

```
docker run -v /host/path:/container/path ubuntu
```

Read-only:

```
docker run -v mydata:/data:ro ubuntu
```

7. Logs & Monitoring

View logs

```
docker logs container_name
```

Flags:

- `-f` → follow (live)
- `--tail 50` → last 50 lines
- `--since 10m`

Resource usage

```
docker stats
```

Shows:

- CPU
- Memory
- Network I/O
- Disk I/O

8. Inspect & Metadata

Inspect object

```
docker inspect container_name
```

Returns:

- IP address
- Mounts
- Env variables
- Network config (JSON)

9. Docker Build (Images)

Build image

```
docker build -t myapp .
```

Flags:

- `-t` → tag image
- `-f Dockerfile.dev` → alternate Dockerfile
- `--no-cache`

Example Dockerfile

```
FROM ubuntu:22.04
RUN apt update && apt install -y nginx
CMD ["nginx", "-g", "daemon off;"]
```

10. Cleanup Commands

Remove unused containers

```
docker container prune
```

Remove unused images

```
docker image prune
```

Remove everything unused

```
docker system prune
```

Flags:

- `-a` → remove all unused images
- `--volumes`

11. Docker Compose (Overview)

Start services

```
docker compose up
```

Flags:

- `-d` → detached
- `--build`

Stop services

```
docker compose down
```

12. Important Run Flags (Quick Reference)

Flag	Meaning
<code>-it</code>	Interactive terminal
<code>-d</code>	Detached mode
<code>--rm</code>	Auto-remove
<code>--name</code>	Container name
<code>-p</code>	Port mapping
<code>-v</code>	Volume mount
<code>-e</code>	Environment variable
<code>--network</code>	Custom network
<code>--restart=always</code>	Auto restart

13. Minimal Lab Example (Ubuntu + Nginx)

```
docker pull nginx
docker run -d --name web -p 8080:80 nginx
docker ps
docker logs web
docker stop web
docker rm web
```

14. Key Concept Summary

- **Image** → Blueprint
- **Container** → Running instance
- **Volume** → Persistent data
- **Network** → Container communication
- **Dockerfile** → Image build instructions